



This is Elena, a typical factory director

At 8:10 on Monday morning, Elena has two hours to make a customer commitment. Sales, production, maintenance and quality all have to be in sync today, as every day.

She is accountable for the whole factory

but...



"Tom says the order's finished. Quality says they're still holding two pallets. So which is it?"

"Anna promised the customer Friday, but Production planned Monday. The teams aren't working from the same plan at all."

"Fernando warned us about a strange vibration yesterday. Night shift never saw the warning. They switched the machine back on."

"Every department says they're on target. Then explain why we've got 200 orders sitting between drying and sorting?"

"Hiroshi retires next month. Thirty years of knowing how this place works, and most of it is still only in his head."



"Carlos can see the energy bill going up. He just can't tell me which machines are causing it."

"Tim's team made the exact same mistake yesterday's shift made. Why didn't anybody tell them?"

"Line Three is down again. The guy who fixed it last time isn't here, and nobody knows what to do."

"Sofia found the defect, but we can't trace which machine or material caused it. The engineers won't come until late this afternoon."

"Day shift made 920. Night shift made 760. Same line, same product, same factory. Nobody can tell me why."

"Elena already told the customer their order was ready. Logistics just told me it's not making today's truck. Why?"

Every system records a piece of the factory. None records the decisions.



EVERY OPERATIONAL PAIN IS DOWNSTREAM OF ONE CAUSE: THE DECISIONS BETWEEN SYSTEMS ARE NEVER RECORDED

Skilled labour
shortage

Aging equipment
breakdowns

Material and cost
leakage

Supply chain blind
spots

Energy wasteful
consumption

Her systems see the plan.

Her floor also runs on unwritten rules.

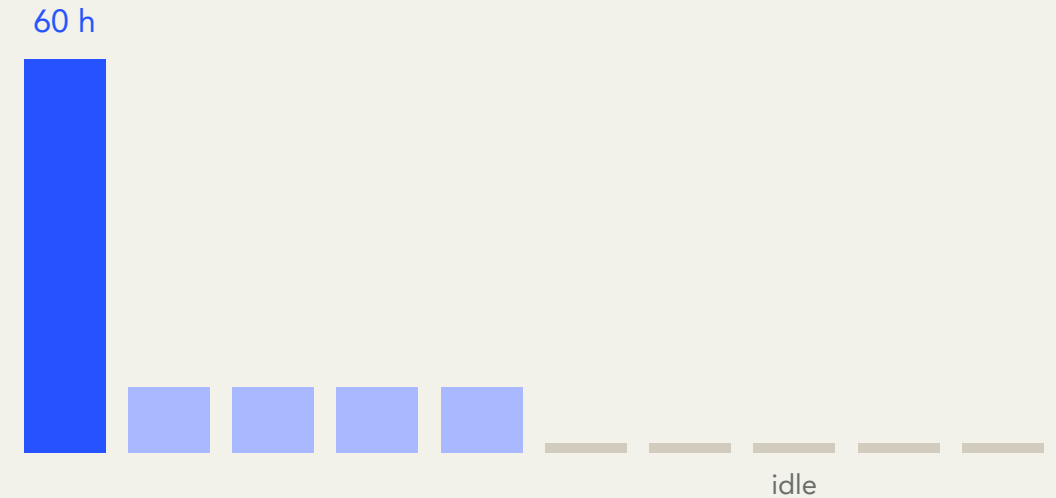
THE PLAN · WHAT YOUR SYSTEM SEES

100 hours of work. 10 people. 10 hours each.



THE FLOOR · WHAT REALLY HAPPENS

One specialist carries 60 hours. Five people wait.

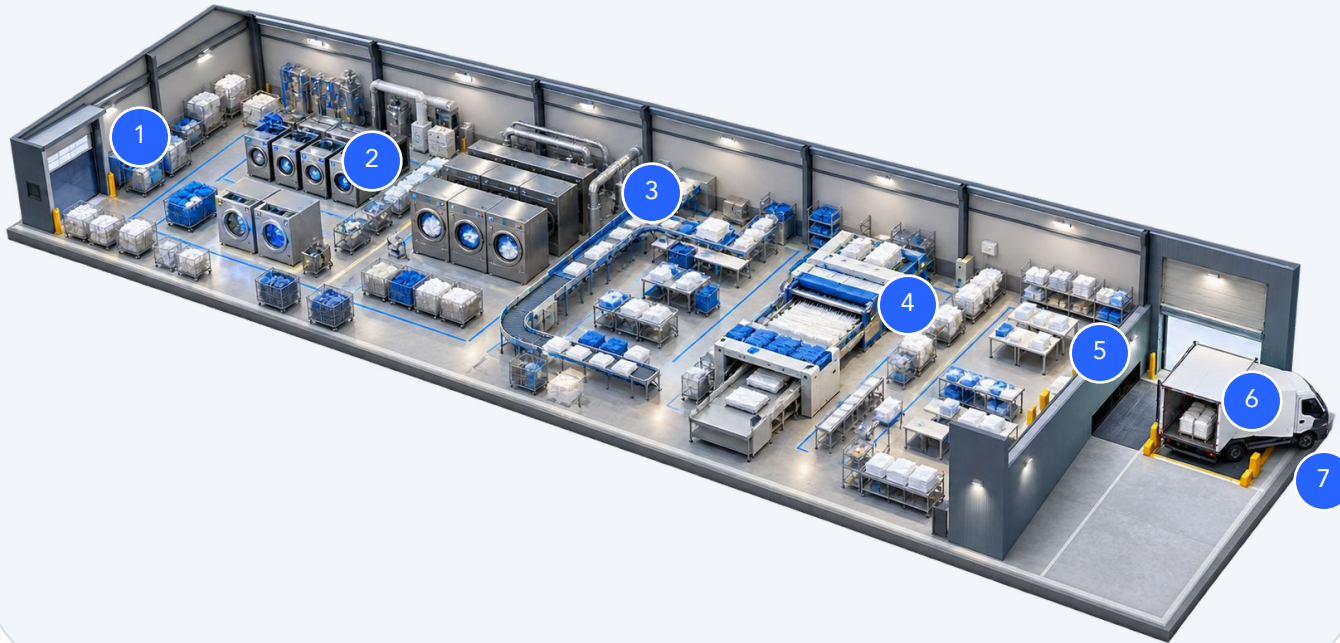


THE GAP BETWEEN THE TWO IS CAPACITY YOU ALREADY PAY FOR. NO PLANT KNOWS ITS REAL CAPACITY.

Context is fragmented across every stage of the factory line

USE CASE: INDUSTRIAL LAUNDRY

ILLUSTRATIVE FACTORY MODEL



WHERE CONTEXT BREAKS

- 1 WAITING**
Priorities are invisible against live capacity.
- 2 WASHING**
Batch, programme, machine state and result stay split.
- 3 DRYING**
Washer output and dryer capacity do not synchronise.
- 4 SORTING**
Order identity, counts and defects lose continuity.
- 5 MACHINE FOLD**
Feed, jams, changeovers and output remain local.
- 6 HAND FOLD + PACK**
WIP, labour and completeness are reconciled manually.
- 7 DELIVERY GATE**
Readiness, routing and customer promise disconnect.

~3,000

ITEMS IN BACKLOG PER DAY

None

LIVE TRACKING AND STATUS

18%

REWASH / REWORK

67%

ON TIME, IN FULL

48 h / wk

MANAGER COORDINATION

The autonomous factory runs itself, with people in command.

LEVEL 1

See

One operational reality across people, machines and systems.

LIVE AT KASA

LEVEL 2

Remember

Every decision, fix and outcome kept as company memory.

LIVE AT KASA

LEVEL 3

Predict

The world model shows what each choice will change.

V0 · NOVEMBER 2026

LEVEL 4

Act

The agent executes through approved tools. People approve what matters.

2027

LEVEL 5

Run

Lines run themselves. Robots take the manual steps.

HORIZON

APPLICATIONS RUN LEVELS 1 AND 2 TODAY AND CAPTURE THE DATA. THE WORLD MODEL TAKES FACTORIES TO LEVELS 3 TO 5.



Celesnity was born 6 months before A16Z posted this video

AI is learning the physical world. Factories are next.

Signals

4x

Physical AI venture funding,
H1 2026 vs H2 2025:
\$47.4bn across 521 deals

40%

Enterprise apps with
built-in AI agents by end of
2026, from under 5% in
2025

5M

Industrial robots now
working in factories;
600,000+ installed in 2025

Breakthroughs

OPEN WORLD MODELS

NVIDIA Cosmos 3 (June 2026)
predicts actions and future states,
and now runs on edge devices. We
fine-tune it.

\$2.3BN IN 6 MONTHS

Invested in world model labs in H1
2026, including about \$1bn each
for AMI Labs and World Labs.

AI NATIVE FACTORIES

Siemens and NVIDIA are building
the first fully AI driven, adaptive
factory in Erlangen, from 2026.

THE INFLECTION: Agents can now act and world models can now predict, but neither has the factory's decisions, which no system records

Minder captures those decisions and trains the world model that lets factories run themselves

ONE SYSTEM,
ONE WORLD MODEL.

From Machines to World Model

Building Industrial Intelligence that Understands Physical Reality

1. MACHINE DATASETS

Raw signals from machines & systems



- PLC / SCADA
- Sensors / IoT
- Logs / Alarms
- Maintenance Records

2. APPLIED ML

ML models applied to real factory data



Anomaly Detection



Predictive Maintenance



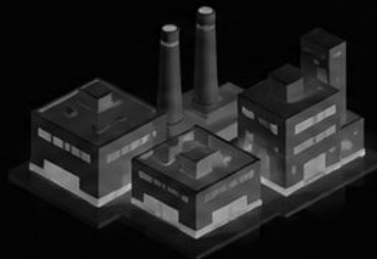
Machine Optimization



Life Cycle Extension

3. REAL DEPLOYMENT

Across verticals, across use cases



Discrete Manufacturing



Process Industry



Energy & Utilities



Food & Beverage



Used by operators, technicians, managers with our AI agent

4. READY DATA

Structured, labeled, and outcome-linked data



Operational Data



Human Decisions & Interventions



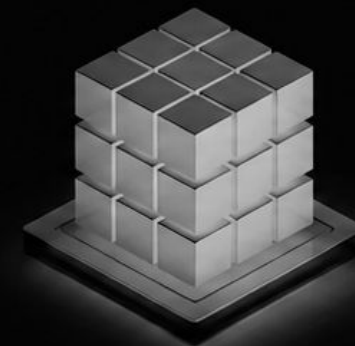
Outcomes & Results



Time & Context

5. WORLD MODEL

A model that understands physical environment, machines & operations



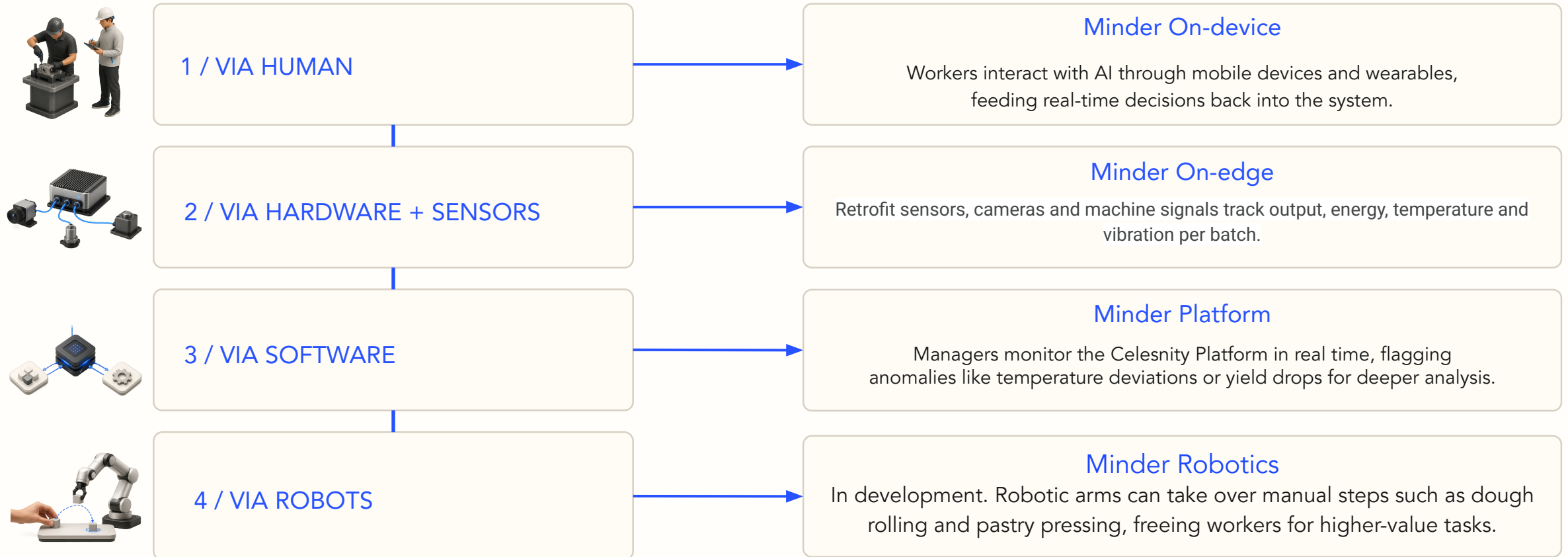
- ✓ Understand physical reality
- ✓ Reason & simulate
- ✓ Predict outcomes
- ✓ Recommend best actions

Continuous Learning Loop

More data → Smarter decisions → Better outcomes → More data

Minder reaches every place where factory work happens.

Case study: food processing, Vietnam



WE MAKE SURE AI REACHES ALL CORNERS OF A FACTORY, INCLUDING DATA-POOR POINTS

Minder adapts to the workers, so the workers never have to learn another system.

Forward Deployed Engineers bridge live workflow
into Minder's intelligence layer

Voice-first. Local language. On existing devices.

OBSERVES

progress and
exceptions

REMEMBERS

procedures, history,
rare fixes

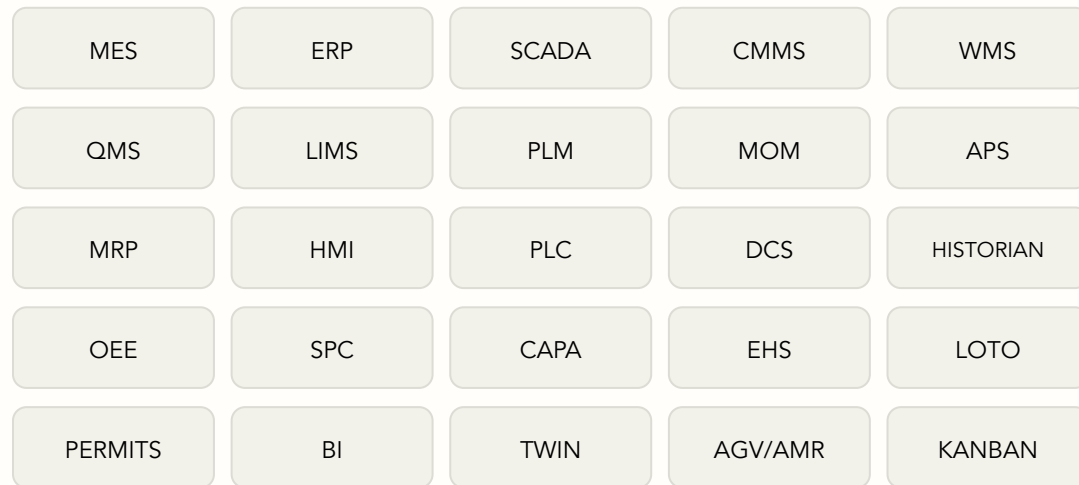
ACTS

through approved
tools



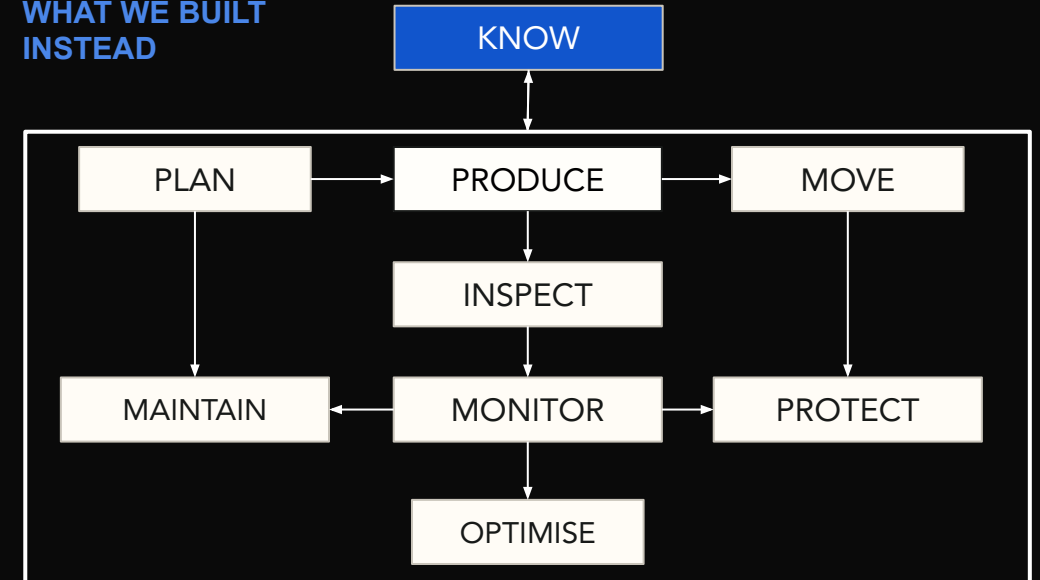
Minder platform fundamentally rethinks how factory data should connect...

WHAT A FACTORY ALREADY RUNS ON (AND STILL FAILED THEM)



+ THE AI COPILOT THAT READS THEM SEPARATELY

WHAT WE BUILT INSTEAD



Existing systems record single pieces of the factory. Factories don't need another AI copilot on top of 20+ siloed systems. Minder records, connects and remembers the whole factory, and every record trains the world model

...so that AI gives Elena & every worker one operational reality

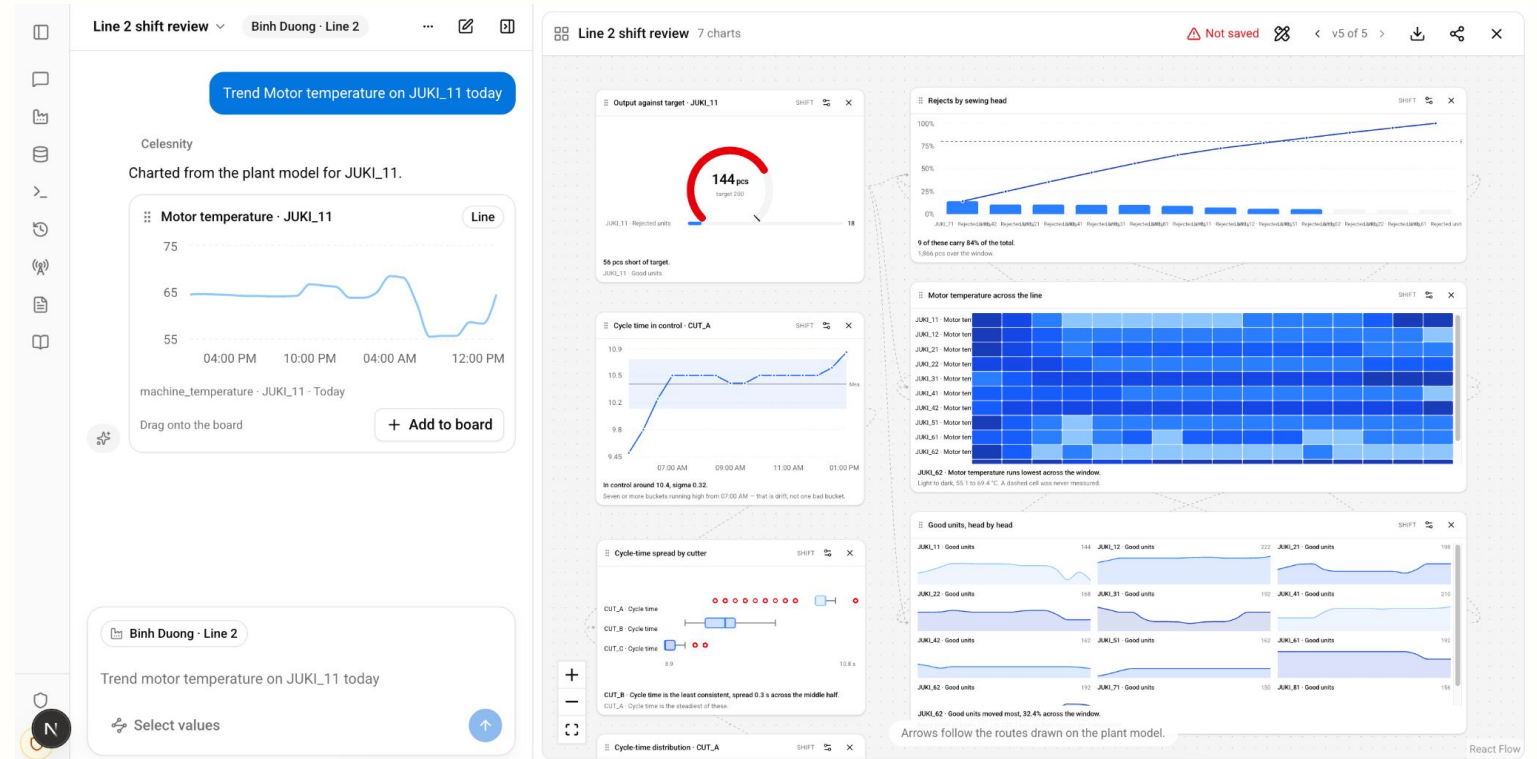
Elena sees what the plant knows, what conflicts and each line's real capacity.

KNOW

The relevant order, work, equipment, people and constraints

DECIDE

The available options, consequences and required approvals



Every worker remains accountable

The missing dataset: what people decide, and what follows.

Minder is designed to capture what robot trajectories usually omit: context, intent, intervention and consequence.

CONVENTIONAL ROBOT TRAJECTORY

OBSERVATION → ACTION → TASK RESULT

Strong at embodiment + motor control

INDUSTRIAL EXPERIENCE TRAJECTORY

CONTEXT

line · product ·
ambient

STATE (t)

signals · cycle ·
alarms

GOAL

task · question ·
KPI

ACTION

command ·
approval

NEXT STATE

measured
response

OUTCOME

quality · time ·
energy

EXPLANATION

reason ·
diagnosis

State grounds language · commands ground actions · telemetry grounds dynamics · outcomes ground value

WHY THIS CAN BE RICHER

It captures why the system changed, what happened next and the experts' rare fixes.

V-JEPA 2

>1M h passive video + <62 h
action-conditioned robot video
supported selected zero-shot planning.

FACTORYNET

51M datapoints / 23K executions; early
cross-embodiment transfer reported.

Elena sees what each choice will change, before she decides.

ELENA ASKS

What if we keep Line Three running two more hours?

RUN AS PLANNED

Order ships. Breakdown
risk rises.

Bearing temperature passes its limit
before the shift ends.

RUN AT 80% · RECOMMENDED

Order ships. Risk stays low.

The slower run keeps the bearing in range
and still makes the truck.

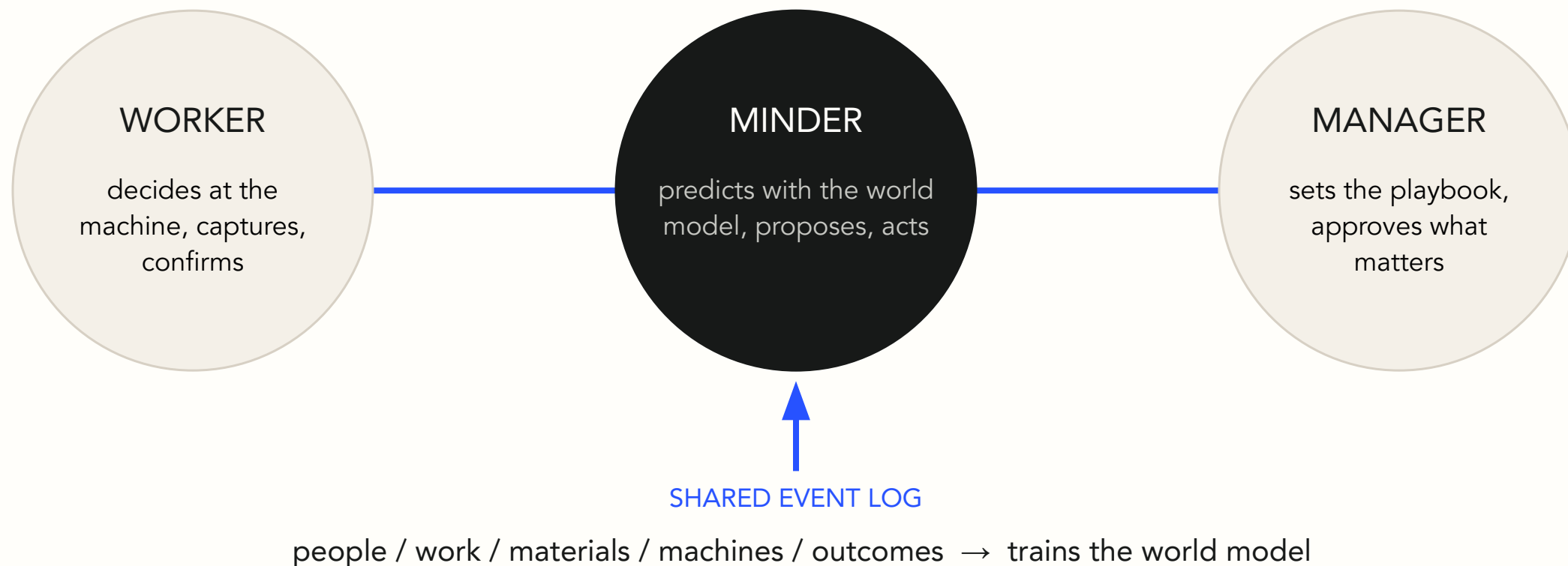
STOP NOW

Order slips a day.

No risk to the machine, but the customer
promise breaks.

Minder does not wait to be asked.

It *proactively* observes, predicts, proposes and acts



At Kasa, field adoption became commercial proof.

Industrial laundry / Phu Quoc / 20 tonnes processed per day / live deployment

7

departments live

100%

assigned worker adoption

6

weeks with
2 FDEs

24

month paid
contract

Measured and agreed with Kasa, results approved by Kasa (April 2026 baseline vs August 2026 results)

-59%

rewash rate

-80%

missing items

85%

on time delivery

30M VND MONTHLY CONTRACT
VALUE

SIGNED / INVOICED
SINCE SEP 2026

FDEs in action!



Hòa Phát, Vietnam's largest manufacturer, is our first world model partner.

NETWORK

~30 factories

Operations data across a national group of steel and appliance plants.

APPLICATIONS

Full suite

Platform and agent deployed plant by plant, on founding terms.

WORLD MODEL

Nov 2026

Celesnity first world model version 0.1

DEPLOYMENT

Their AI team

Trained by us as FDEs, paid by Hòa Phát.

THEY GET EARLY ACCESS AND FOUNDING TERMS. WE KEEP THE WEIGHTS, CODE AND TRAINING RIGHTS. IN DISCUSSION: NESTLÉ, ISUZU.

Demand is emerging across nine industrial contexts.

\$6.49bn

STAGE MIX

1 PAID

8 MOU SIGNED

3 SCOPING

8 PROSPECTING

~35000 employees

Total combined revenue of
current partners in pilots and
conversations

- Kasa Laundry
Industrial laundry

- HQA Laundry
Industrial laundry
- Hòa Phát Group
Steel and appliances
- Saigon Food
Food processing
- Probiotics
Biotech
- City Water Tan Hiep
Water
- Hue One Food
Food processing
- Hakawa Laundry
Industrial laundry
- Van Minh Huy Laundry
Industrial laundry

- Nestlé Vietnam
Food processing
- SCG Vietnam
Cement
- Isuzu Vietnam
Mobility manufacturing

- Chan May Port
Port
- City Water Thu Duc
Water
- Scavi
Textiles
- Expand Vietnam
Food processing
- Five Stars Laundry
Industrial laundry

Every live site feeds one world model. Industrial groups join as founding world model partners.

All industrial AI reads the factory's software. Minder learns the factory itself.

CATEGORY	BEGINS WITH	LEARNS FROM
INDUSTRIAL SUITES <i>Siemens · AVEVA</i>	installed enterprise estate	machine and system records
FRONTLINE PLATFORMS <i>Tulip · Augmentir</i>	configured apps and workflows	forms workers fill in
AI COPILOTS <i>Cognite · Augury · MaintainX</i>	existing data and context	whatever the stack already holds
MINDER AI	voice-first worker adoption	decisions, actions and outcomes, captured at source

Competitors learn from the data their stack already holds. Minder creates the data no stack records.

\$100bn a year to win in three regions (China excluded)

\$100bn

TAM a year: 10% more capacity across ~\$5tn of manufacturing output in the three regions, with Minder capturing 20% of the value.

Plus the World Model API, not yet sized. China excluded.

\$390m

SOM within 5–7 years: about 2,500 plants, 1% of the 246,600 we can serve, starting with Hòa Phát’s 30 factories.

SAM per plant	Southeast Asia	Europe, East Asia
Mid plant	\$67k	\$135k
Large plant	\$225k	\$449k

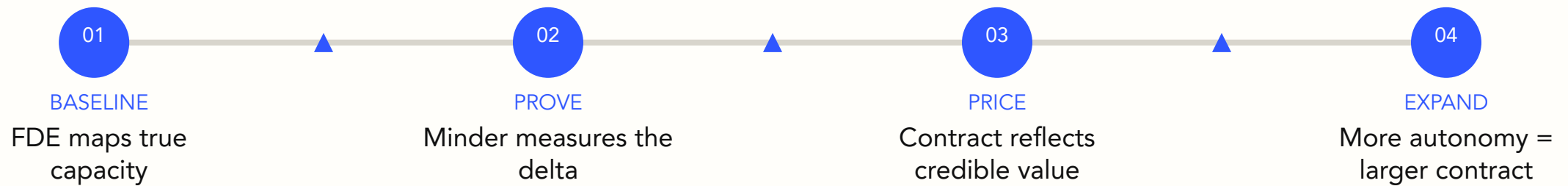
SAM \$39bn

Applications \$26bn + world model \$13bn



All figures a year, in US\$ at \$1.30 per £. China excluded. TAM is an assumption-led value pool: 10% more capacity on ~\$5tn of manufacturing output in Europe, East Asia and Southeast Asia, 20% captured. SAM: 246,600 plants with 50+ employees (Europe 128,000 · \$22.8bn; East Asia 56,400 · \$10.0bn; Southeast Asia 62,200 · \$6.5bn). EU from Eurostat SBS 2023 enterprise counts converted to plants; other markets estimated from official national totals (ONS, METI, KOSIS, DGBAS, BPS, DOSM) or comparable markets. Per plant: applications at operating model list (\$150k per large plant in Southeast Asia, 2x in Europe and East Asia, mid plants at 30%) plus a world model licence at 50%. SOM: about 1% of SAM plants.

Minder charges a % of the value we create



Starting

Find the real capacity of one line



Expansion

Baseline + attribution + measured outcomes



Applications + world model

One operational world model of industry, sold to everyone who runs, equips, connects, automates, insures or governs it.

THE UNIQUE DATA ASSET

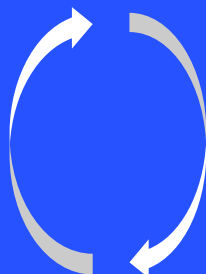
Worker behaviours
decisions and tacit know-how

Asset vision
what physically happened

Machine telemetry
how equipment really behaves

+ Outcomes

Compounds with every site vertically and horizontally



01 Factories

Minder runs on it: forecasts, planning and governed autonomy

Site licence + savings share

02 Machine builders

Ships inside new lines, compressors and CNC machines

Per machine royalty

03 Software vendors

API that makes MES, CMMS and ERP stacks predictive

Licence + API usage

04 Robot makers

Real factory context to train and test robots

Dataset + simulation fees

05 Insurers & lenders

Asset risk priced on how machines actually run

Per asset risk subscription

06 Governments

National industrial intelligence and productivity

Multi-year sovereign deals

THE PRIZE

\$16.8T

manufacturing value added a year

~\$5T

of it in Europe, East Asia and Southeast Asia, China excl

~\$50B

of value a year for every 1% of capacity unlocked there

WORLD MODELS ARE ALREADY PRICED IN BILLIONS

\$4.5B

AMI Labs

~\$5B

World Labs

\$11B+

Physical Intelligence

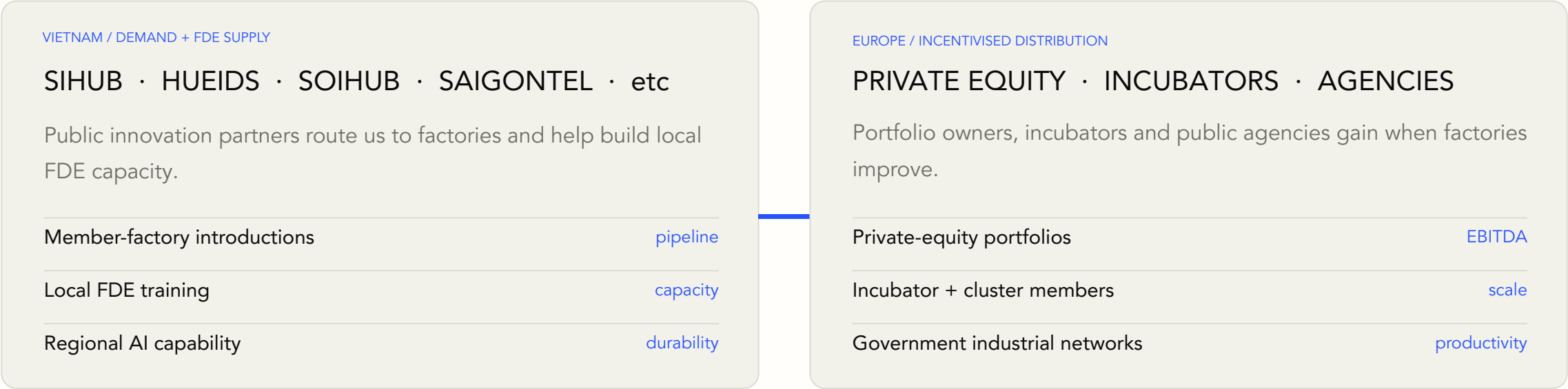
\$14B+

Skind AI

Sources: World Bank, manufacturing value added 2024. Regional share (~\$5T, China excluded) is a Celesnity estimate. 2026 valuations per Bloomberg, TechCrunch and company releases (World Labs, Physical Intelligence as reported). ~\$50B is illustrative value, not contracted revenue.

Partners generate our pipeline.

We recruit channels whose economics improve when their factories improve. Their incentives make introductions natural. Midsize plants come through channels. Industrial groups join as founding world model partners, and their AI teams become our FDEs.



Owned, hosted and governed in Europe.

Our Vietnam subsidiary gives us an engineering cost and speed advantage. But Europe holds the company, the IP, the data and the controls.



OWNERSHIP	Celesnity Ltd, London UK, owns all IP.
DATA RESIDENCY	EU data stays in EU regions, UK data in London. Edge mode keeps plant data on site.
DATA ACCESS	Only European staff touch customer data. Vietnam builds on code and anonymised data.
DATA RIGHTS	Customers own their raw data. World model contribution is opt-in and anonymised.
MODELS	Model agnostic. In-region inference, with an open-weight option on customer premises.
WORKERS	No individual worker scoring. GDPR and works council ready. AI Act controls by Dec 2027.



**We do not build for factories from behind a desk.
We build with them, inside them.**



Built by people who can make it happen.

CO-FOUNDER / CEO

Eric Nguyen

Founders for Founders s16vc + AltaIR Capital AltaLab

Field deployment, commercial motion and partnerships.
Led the Kasa deployment in 6 weeks, founded Strand Ventures (UK's first student VC).

B2B GTM / finance / investor relations / operation / growth

CO-FOUNDER / CTO

Bill Nguyen

Silicon Valley Fellow + Founders for Founders s16vc

Built an open-source AI agent for Snapchat Spectacles smart glasses. Built Jan v3.5 (400k downloads) & Byterover CLI (5k starred)

AI research / product / users / open source / engineering

OUR TEAM OF 21, FOUNDERS INCLUDED



\$5M

raising now

EIS-eligible; HMRC Advance Assurance in preparation. UK topco: Celesnity Ltd.

to ship the industrial world model and run it in 50 to 100 partner plants by 2028

MILESTONES THE CAPITAL BUYS

Nov 2026	first model tested at a Hòa Phát plant
Mar 2027	first model live on 5 more plants, beating their baselines
Dec 2027	world model across at least 20 plants
Sep 2028	world model running at 50 to 100 plants

USE OF FUNDS · 24 MONTHS

World model team and evaluation	29.1%
Platform, agent and inference	22.1%
G&A, security and legal	15.1%
GPU compute and data, after credits	15.0%
Forward deployment	11.0%
Go to market	7.7%

World model team, compute and data are 44% of planned spend.
Up to \$500k of Google Cloud and AWS credits cover early training.

£100k angel, otherwise bootstrapped



The next morning, Elena still owns the decision. Minder gives her the understanding to make it with confidence.

We are raising \$5M to ship the first industrial world model, with our applications, into a \$100bn a year opportunity, starting with Hòa Phát.

Back the autonomous factory, built on the world model of industry.



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